Homework

DM10 - Points of a tetrahedron - 50 min.

Consider a tetrahedron $ABCD$ not flat. The points I, J and K are the respective midpoints of $[AB]$, $[AC]$ and $[AD]$.

1. Express $\overline{IJ}$ as a function of $\overrightarrow{BC}$ and $\overrightarrow{IK}$ as a function of $\overrightarrow{BD}$.
2. Deduce that the planes (IJK) and (BCD) are parallel.
3. Define the point G by $\overrightarrow{AG} = \frac{1}{4}(\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD})$. The point G lie in the plane (IJK) ? Justify.
4. For this question only, assume that, in an orthonormal coordinate system, $A(0;0;0)$, $B(4;0;0)$, $C(0;4;0)$ and $D(0;0;4)$. Calculate the lengths IJ, IK and JK.